

Hardware Processing Engines on FPGA

Evaluating open-source RISC-V platforms for Edge AI accelerators

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Greetings everyone! I am going to present my findings from my UGRC under Dr. Karthik, prof Nitin and prof Gopal. I have been researching hardware accelerators for the past 4 months and I am excited to share some of the results of my study.

Motivation: Edge AI is power- and area-constrained

The field of AI has had several breakthroughs in the past decade leading to widespread adoption in almost every field. Many such applications are constrained by power($\approx$ milliwatts) and area($\approx$ square millimeters) of the inference unit.

Examples:

- **Rail fault detection** — on-track, battery-powered units.
- **Breast cancer screening** — handheld imaging devices.
- **Driver assistance (ADAS)** — latency bound.

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AI has taken off. Industries are trying to integrate AI into their systems. The proliferation of AI has been accelerating and shows no signs of slowing down. This leads to a very interesting problem. There is obviously no one-size-fits-all solution to deploying AI. The field of Edge AI looks at applications where power and area are the biggest constraints.

There are around 66 million pieces of academia related to Edge AI on Google Scholar!

Four constraints define the design space

- 1. **Power.** A few hundred milliwatts in total
- 2. **Area.**
- 3. **Extensibility.** Allow for custom workloads and optimization of these workloads.
- 4. **Toolchain.** A C compiler that can emit instructions for the hardware.

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We start by defining the constraints of our study. As I said before, we are in the Edge AI domain so power and area are the biggest constraints. On top of that, we require platforms to be extensible(because applications have varying constraints, extensibility is very crucial, platforms amenable to change are better). A toolchain is essential for testing software programs right out of the box. We also mandate that platforms be fully open-source and based on RISC-V.

Why these constraints you may ask? We want to keep things generic but specify enough to limit the scope of the study.

Edge-AI hardware is fragmented across every axis

A unified platform would have to span the following combinatorial space:

CPU cores	RI5CY / CV32E40P, CV32E40PX, CVA6, Snitch, Ibex, Rocket / BOOM, OpenC910, Arm Cortex-M, Tensilica Xtenna
Accelerators	Quadrilatero, Gemmini, RedMuIE, NE16 / NEUREKA, ITA, VTA, Coral NPU, Hailo-8, Ethos-U, custom HWPEs
Accel. interfaces	CV-X-IF, HWPE-Stream, RoCC, SSR / FREP, AXI / AXI-Stream, OBI, TileLink, vendor-private
FPGA targets	Nexys-A7 (Artix-7), Genesys2 (Kintex-7), ZCU102 (UltraScale+), VCU118, PYNQ-Z2, Arty, Intel DE-series, Lattice ECP5
Compilers	GCC (riscv-gnu), LLVM / Clang, xheep_matrix_spec LLVM, vendor SDKs, TFLite Micro, TVM, Exo, Deeploy, MLIR
Simulators / EDA	Verilator, QuestaSim, VCS, Vivado / Vitis, GVSoC, Spike, QEMU, Renode, FuseSoC, Bender

Cores × *Accels* × *Interfaces* × *FPGAs* × *Compilers* × *Simulators* — no single platform spans it.

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Edge-AI hardware is fragmented across every axis

Even after all those constraints, there are many options to choose from. A researcher trying to implement his application would have to choose from

CPU core, accelerator This is a very daunting task unless there is a guideline on what to choose for a particular class of applications. That is the

core motivation of our study. How can we give the researcher a head start in choosing a particular framework for his application.

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Part I.

The platform survey

Platform 1: Coral NPU

Evaluated as a candidate for rapid edge-AI prototyping (Feb 2026).

Issues.

- No FPGA flow.
- No RISC-V Matrix core.

Verdict. Abandoned. Pivoted to the open-source PULP ecosystem.

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We first looked at Coral NPU. There is no support for FPGA. Coral NPU is still under development. A RISC-V vector core has been added and a matrix core is in development. Add to that, the software was restricted to Google's LiteRT/Tensorflow framework.

Platform 2: PULPissimo — toolchain is too old

Hardware is sound. The software around it has gone end-of-life.

Issues.

- ▶ SDK requires gcc-5 (only Ubuntu 18); fails on modern hosts.
- ▶ RTL simulation requires QuestaSim/Xcelium(not open-source); ModelSim trial fails on common_cells(not officially supported).
- ▶ Supported FPGAs not available in RISE Lab.

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$ make build
make[2]: Leaving directory '/home/Myuser/riscv/pulp/rtl/tb/remote_bitbang'
** Warning: ../ips/common_cells/src/fifo_v3.sv(13): (vlog-2186)
SystemVerilog testbench feature
(randomization, coverage or assertion) detected in the design.
These features are only supported in Questasim.
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Verdict. Abandoned — reproducibility unacceptable for developing applications.

Platform 2: PULPissimo — toolchain is too old

We shifted focus to PULP. PULP is an ecosystem. There are many platforms within PULP that together offer great performance over a range of power constraints.
The great idea is, Use many many cores that are very very efficient.

PULPissimo is a single-core platform.. The PULPissimo architecture includes: Either the RI5CY core or the Ibex one as main core Autonomous Input/Output subsystem (uDMA) New memory subsystem Support for Hardware Processing Engines (HWPEs) New simple interrupt controller New peripherals

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Platform 3: PULP-derived alternatives

Platform	Blocker	Status
Chimera	Documentation thin; no FPGA flow.	rejected
Siracusa	No public repository.	rejected
Mempool + ITA	FPGA infeasible at full scale.	rejected
Snitch Cluster	No FPGA support out of the box.	rejected
GAP9	Proprietary; not open-source.	rejected
Cheshire	FPGA flow exists; no Deeploy backend.	set aside

Common limitation: no ready-made FPGA synthesis flow combined with an accelerator-integration pipeline.

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The last two platforms were limited by software support. Thus we focussed on a software toolchain and tried to find hardware platforms that would support the toolchain. We found Deeploy by PULP.

- Deeploy is an ONNX-to-C compiler that generates low-level optimized C Code for multi-cluster, heterogeneous SoCs. Its goal is to enable configurable deployment flows from a bottom-up compiler perspective, modeling target hardware in a fine-grained and modular manner. Chimera is an open-source, highly configurable microcontroller System-on-Chip (SoC) template designed for multi-cluster, heterogeneous computing systems. Its primary objective is to provide a modular platform for seamlessly integrating and managing hardware accelerators, offering developers and researchers an intuitive and extensible foundation.
- MemPool is a many-core system targeting image processing and wireless applications. It implements 256 RISC-V cores that can access a large, shared L1 memory in at most five cycles. TeraPool and MinPool, respectively a 1024 RISC-V cores scaled-up and a 16 RISC-V cores scaled-down parametrizations of MemPool are also supported.
- The RISC-V Snitch integer core. This can be useful stand-alone if you are just interested in re-using the core for your project, e.g., as a tiny control core or you want to make a peripheral smart. The sky is the limit. The Snitch cluster. A highly configurable cluster containing one to many integer cores with optional floating-point capabilities as well as our custom ISA extensions Xssr, Xfrep, and Xdma. A runtime and example applications for the Snitch cluster. RTL simulation environments for Verilator, Questa Advanced Simulator, and VCS, as well as configurations for the GVSoC system simulator.
- Cheshire is a minimal Linux-capable host platform built around the RISC-V CVA6 core. Its goal is to provide a lightweight, configurable, autonomously booting host to systems that need one, from minimal Linux-capable SoCs to manycore compute accelerators.

Platform 4: X-HEEP + Quadrilatero — adopted

X-HEEP is a configurable RISC-V microcontroller (by EPFL-ESL) derived from the PULP ecosystem.

Quadrilatero is a systolic-array co-processor. Communicate over CV-X-IF.

- **FPGA-first.** First-class support for Nexys-A7 and Genesys2.
- **Active toolchain.** Docker image, Makefile flow, Verilator-support.
- **CV-X-IF interface.** Co-processors attach without modifying MCU RTL. Incredible eXtensibility.
- **Quadrilatero.** Open systolic array implementing the custom `xtheadmatrix` ISA extension.

Confirmed as the project target after consultation with Dr. Karthik.

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Part II.

Bringing up the system

System Architecture

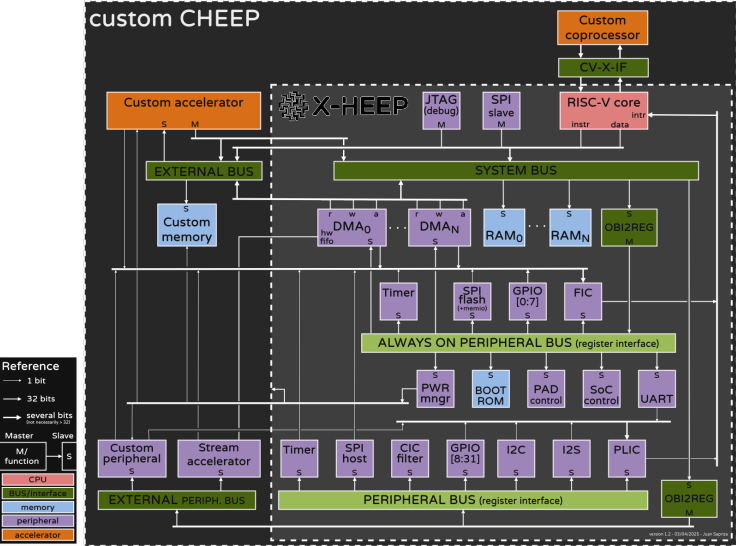


Figure: X-HEEP Microcontroller

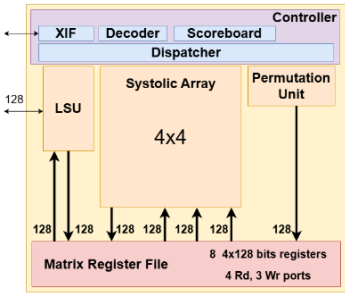
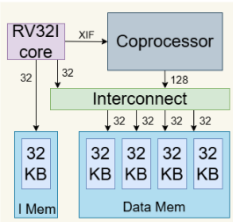


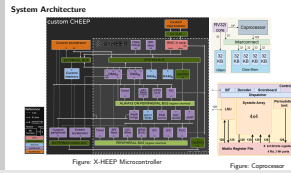
Figure: Coprocessor

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System Architecture

X-HEEP is a standard microcontroller with peripheral support. Quadrilatero is a 4x4 systolic array coprocessor. Both connected through CV-X-IF. Quadrilatero has its own LSU so it can load and store into the the main processors RAM independently. The exact memory and peripherals can be configured easily using python.



System Architecture

- ▶ **CPU:** CV32E40PX, a 32-bit 4-stage inorder processor developed by PULP.
- ▶ **Coprocessor:** Quadrilatero, A 4x4 systolic array co-processor.
- ▶ **Interface:** CV-X-IF, a RISC-V extension interface to facilitate communication between a CPU and a coprocessor.

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System Architecture

there is not much difference between an coprocessor and an accelerator here since quadrilatero already has an LSU with direct connection to memory. It is called a coprocessor because of the interface it uses to communicate with the CPU.

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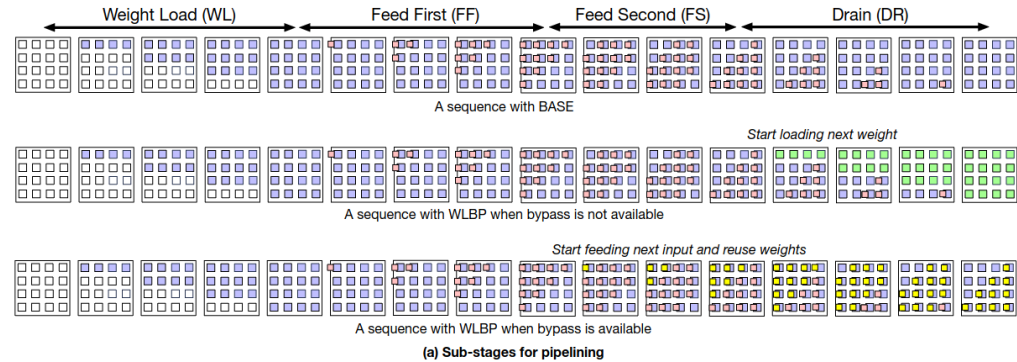
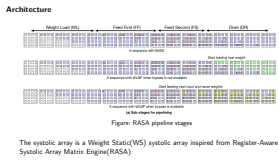


Figure: RASA pipeline stages

The systolic array is a Weight Static(W_S) systolic array inspired from Register-Aware Systolic Array Matrix Engine(RASA).

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Architecture



Each instruction comes through the XIF interface. It is then decoded and passed to three possible units by the dispatcher. mzero goes to the permutation unit which sets all registers to zero. mld goes to the LSU and mmasa goes to the systolic array. WL: The systolic array loads the operands from the matrix register file using the scoreboard to prevent data races with the LSU and permutation unit. The scoreboard essentially maintains whether a particular register can be read or not. FF: feed first. FS: feed second. DR: drain stage.

Layer 1: FPGA bitstream

Custom Dockerfile on top of x-heep-toolchain, with Vivado *mounted* as an external volume rather than bundled.

Issue — Vivado version. The FPGA scripts use `-freq_hz` in `create_bd_port`, introduced after 2018.3.

```
Unknown option '-freq_hz', please type 'create_bd_port -help' for usage info.
while executing "source xilinx_generate_clk_wizard.tcl"
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Resolution. Mount Vivado 2022.1+; then:

```
$ make mcu-gen
$ make vivado-fpga FPGA_BOARD=nexys-a7-200t
```

produces a complete bitstream.

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Layer 1: FPGA bitstream

The first goal was to test FPGA support since all platforms failed this constraint. This worked out of the box with Vivado 2021 and higher.

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Layers 2 & 3: Simulator and compiler

Verilator (COMBDLY). Verilator 5.x escalates a non-blocking-in-comb warning to an error; found in CV32E40PX clock-gate model. Fix:

```
/* verilator lint_off COMBDLY */
if (clk_i == 1'b0) clk_en <= en_i | scan_cg_en_i;
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LLVM (xtheadmatrix). The extension is not in upstream LLVM; build the patched compiler from xheep_matrix_spec and install into a user-owned mounted directory (and not /usr due to ownership issues):

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Layers 2 & 3: Simulator and compiler

Next I tried to compile the hardware and software application for simulation. This worked with a few fixes.

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Part III.

The Quadrilatero bug

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The Quadrilatero bug

But running a matrix multiplication example led to a wrong output or a deadlock. I investigated this issue further.

Symptom: result-valid never asserted

Compiled example_matmul_quadilatero, simulated under Verilator, inspected waveforms in GTKWave.

Observed:

- 1. CPU dispatches every xtheadmatrix instruction over CV-X-IF. ✓
- 2. Quadilatero accepts each instruction (accept = 1). ✓
- 3. *Result-valid is never asserted.* ✗

This rules out interface-level errors. The issue lies inside Quadilatero.

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Root cause: valid-ready deadlock; both sides waited

Before — deadlock.

- ▶ RF sequencer: waited for ready before asserting valid.
- ▶ Systolic array: waited for valid before asserting ready.

Underlying cause: the W, A, D registers load from memory with non-zero latency; the original code held valid low until all three landed.

Fix.

- ▶ RF sequencer asserts valid per individual register, as each loads.
- ▶ Systolic array asserts ready only once *all three* (W, A, D) have arrived.

ready-valid handshake semantics fixed. Correct results for matrix inputs up to 16×16; 32×32 works after relinking at 0x10000; 64×64 still constrained by the SRAM bank configuration.

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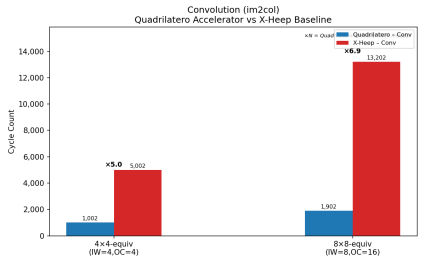
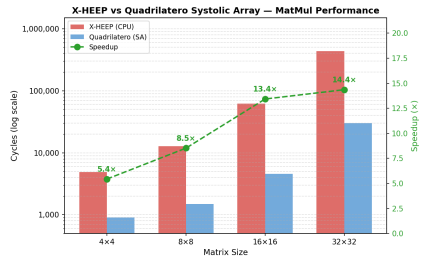
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Part IV.

Conclusion and Future work

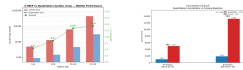
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Delivered.

- ▶ Reproducible Docker build environment.
- ▶ Working Verilator simulation; GEMM and convolution verified.
- ▶ RTL fix for the Quadrilatero handshake deadlock.
- ▶ Generated bitstream for the Nexys-A7-200t.

Next.

- ▶ FPGA validation against the simulation.
- ▶ Deeploy → `xtheadmatrix` integration.

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Lessons Learned

- ▶ Software is the **bottleneck**.
- ▶ High barrier of entry for engineers trying to solve **real-world** problems.
- ▶ Dependency hell is **real**.
- ▶ Docker containers **rock**.
- ▶ Research in Edge AI should be **easy** to reproduce.

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- Software is the **bottleneck**. Writing modular compilers that can support custom instructions. Software pipelines for FPGA, ASICs are mostly figured out but widespread adoption of a single flow likely not happening. A good edge AI SDK would support these industry standard pipelines instead of choosing one and forcing it on the user. On the software side, the abstraction is still not there yet.
- High barrier of entry for engineers trying to solve **real-world** problems. Look at the number of platforms they offer and the number of platforms that have any documentation for testing on an FPGA. The research is fast moving, not interested in making their platform commercially viable yet. May change in the future.
- Dependency hell is **real**. This is perhaps the curse that plagues all of software. In PULP’s case though, the ecosystem’s software framework was last updated in 2021. Since then, 3 versions of Ubuntu have been released. Most required dependencies of PULP software are out of commision in Ubuntu 24.04 and even 22.04.
- Docker containers **rock**. Solve the dependency hell issues but we are missing the issue here. This is a problem we caused in the first place! Anything on top trying to fix it is just shoddy patch job to not break things. Alas, such are the dynamics in the OSS world.
- Research in Edge AI should be **easy** to reproduce. This field lies at the fence of pure research and engineering. At the edge. Thus special focus should be paid to make the research as reproducible as possible. A bug in the RTL is last thing someone would expect when the accelerator they were testing from a research paper does not work. Software should be developed such that it works on all machines if it works on one.

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